Flipkart

Descriptive Analytics Report: Flipkart

Daily Purchase-Transaction Behavior at Scale

Tracking Purchase Trends, Product Categories, Regional Demand & Customer Behavior

Note on data: This report uses a Flipkart simulated purchase-transaction dataset containing 15,425 transactions across 90 dates (1 March – 29 May 2026), 14 cities and 5 product categories. The dataset is row-level transaction data rather than pre-aggregated summaries. Therefore, the analysis derives all KPIs directly from the supplied records and does not assume a larger real-time Flipkart transaction stream.

Keywords

Core themes and analytical dimensions covered throughout this descriptive analytics report:

- Flipkart
- Descriptive Analytics
- E-commerce Transactions
- Gross Merchandise Value (GMV)
- Transaction Volume Trends
- Product Category Performance
- Regional / City-wise Demand
- Repeat vs New Customers
- Discount & Pricing Strategy
- Customer Rating (CSAT)
- Weekday vs Weekend Behavior
- Average Order Value (AOV)
- Sale-Event / Festive Spike Analysis
- Category x City Cross-Analysis
- Data-Driven Business Insights
- Predictive Analytics Roadmap

These keywords anchor each analytical section of the report — from transaction-level trends and category economics to regional demand patterns and customer-experience signals — and map directly onto the KPIs summarised on the Executive Summary page.

1. Objective

Build a descriptive analytics layer over Flipkart purchase-transaction data to answer the following recurring business questions:

- How are daily transaction volumes trending over time?
- How is daily GMV trending over time?
- Which product category contributes the highest transaction volume?
- Which product categories contribute the most to overall GMV?
- Which cities generate the highest transaction volumes?
- Which cities contribute the most GMV?
- Average discount levels across categories.
- City-level transaction demand geographically.
- How does repeat-customer share relate to transaction volume?
- Weekday and weekend transaction patterns.
- Customer Rating (equivalent to Customer Satisfaction).

2. How This Would Work at Real Scale

Processing large-scale e-commerce transaction events is an engineering problem before it is an analytics problem. A realistic architecture would collect transaction events, aggregate them by time/category/city, store raw and aggregated data, and expose the resulting KPIs through a data visualization layer.

Layer	Purpose	Typical tools
Investigation	Research, understand and prepare the transaction dataset	Google, Claude & ChatGPT
Stream processing	Aggregate by category, city and time	Jupyter Notebook (Python)
Storage	Retain raw and processed data	Excel files / CSV files
Batch aggregation	Daily KPI and summary metrics	Jupyter Notebook
Report	Present analysis, graphs and insights	Data Visualization / MS Word

3. Daily Transaction Volume

Over the supplied 90-day window, the dataset records 15,425 transactions and approximately Rs 2.11 Cr in GMV. The daily chart below shows how transaction volume moves across the available dates.

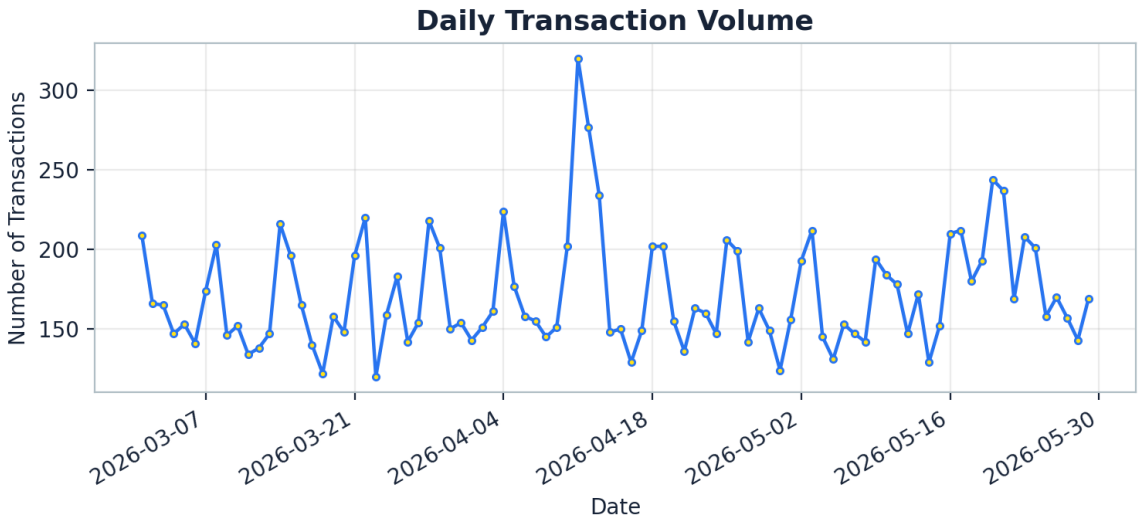


Figure 1. Daily transaction volume

INSIGHTS:

- Daily transaction counts fluctuate broadly across the three-month window, with visible spikes around promotional windows in mid-April and mid-May.
- Higher-volume dates indicate periods of stronger purchase activity, consistent with sale-event demand.
- Monitoring daily transaction volume can help identify demand fluctuations.
- This can support operational and capacity planning.

Total transactions: 15,425

Days covered: 90

Average transactions/day: ~171

4. Daily GMV Trending Over Time

Total GMV in the supplied records is Rs 2.11 Crore (Rs 21,144,007.16). The average GMV per day is ~Rs 234,933 while the highest daily aggregate GMV is ~Rs 435,150. GMV is calculated as the sum of the dataset's order_value_inr field.

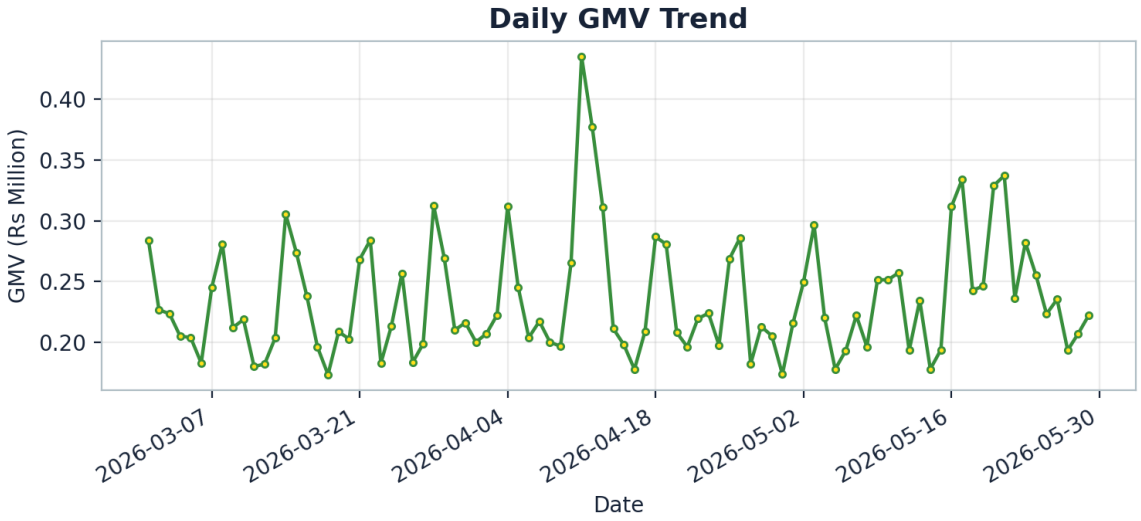


Figure 2. Daily GMV trend

INSIGHTS:

- GMV varies across the available dates, broadly tracking transaction volume.
- Higher GMV indicates stronger monetary transaction activity on that date.
- Daily GMV monitoring helps identify high-value transaction periods, including sale-event spikes.
- GMV should be considered alongside transaction volume to understand demand more completely.

Total GMV: Rs 21,144,007.16

Average GMV per day: ~Rs 234,933

Peak daily GMV: ~Rs 435,150

5. Best-Performing Product Categories

The supplied dataset spans 5 product categories: Electronics, Fashion, Home & Furniture, Grocery & Essentials, and Beauty & Toys. Electronics is the leading category by transaction volume.

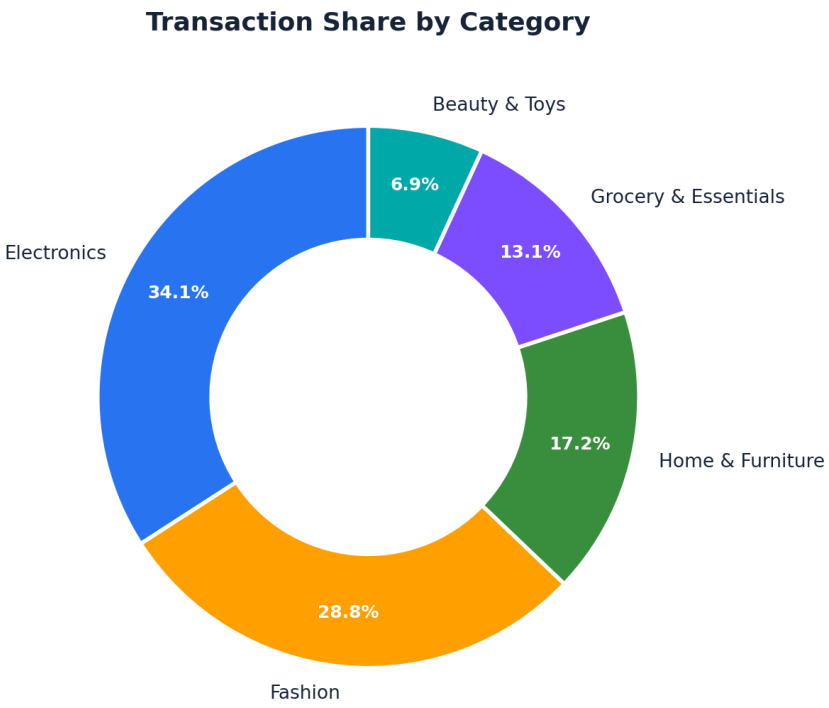


Figure 3. Transaction share by product category

INSIGHTS:

- Electronics leads with 34.11% of all transactions, followed by Fashion at 28.77%.
- Home & Furniture (17.18%), Grocery & Essentials (13.05%), Beauty & Toys (6.88%) make up the remainder.
- The category mix suggests Electronics and Fashion drive the bulk of purchase frequency.

Top category by transactions: Electronics

Transaction share: 34.11%

Categories represented: 5

6. Product Categories Contributing the Most to Overall GMV

While Electronics leads in transaction count, its higher average order value amplifies its GMV contribution even further.

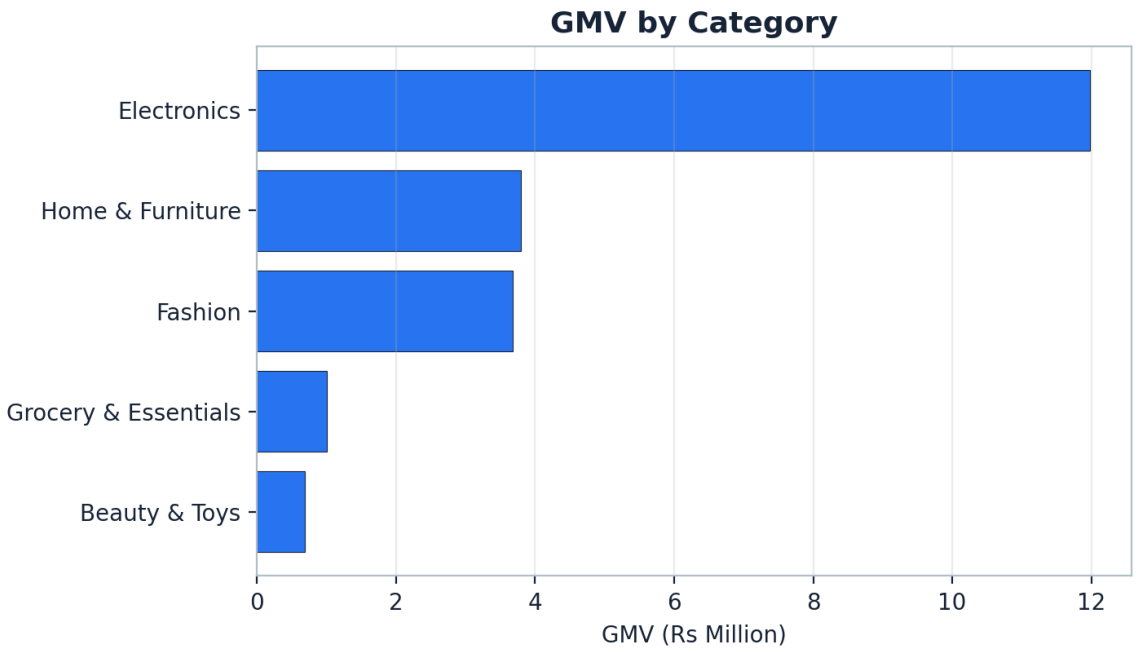


Figure 4. GMV by product category

INSIGHTS:

- Electronics contributes 56.65% of total GMV (Rs 1.20 Cr) despite being 34.11% of transactions — its average order value pulls GMV share well above its transaction share.
- Home & Furniture contributes 17.97% of GMV, and Fashion 17.38%.
- Grocery & Essentials and Beauty & Toys together contribute under 10% of GMV, reflecting their lower price points.

Top GMV category: Electronics

GMV share: 56.65%

Electronics GMV: Rs 11,978,410

7. Average Order Value by Category

Average Order Value (AOV) explains why transaction share and GMV share diverge across categories. The overall platform AOV is Rs 1,371, but individual categories sit well above or below that line.

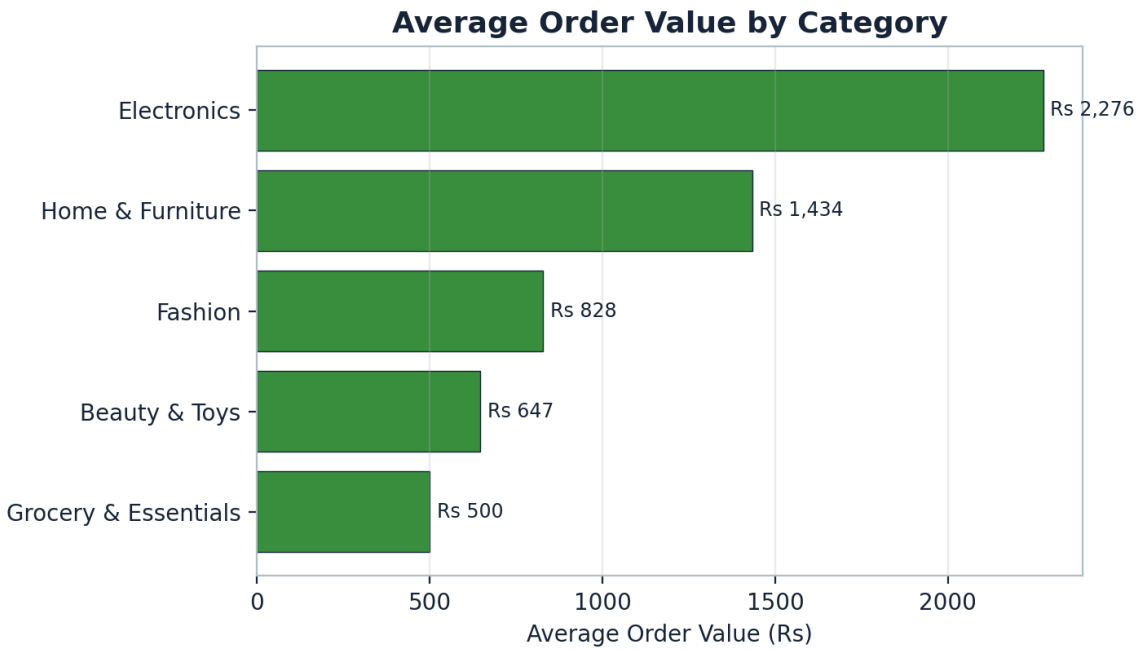


Figure 12. Average order value by category

INSIGHTS:

- Electronics has the highest AOV at Rs 2,276 — roughly 4.6x the AOV of Grocery & Essentials (Rs 500), explaining its outsized GMV contribution.
- Grocery & Essentials carries the lowest AOV, consistent with high-frequency, low-ticket essential purchases rather than considered, big-ticket buys.
- Category-level AOV is a useful lens for pricing, bundling and cross-sell strategy — low-AOV categories are natural candidates for basket-building promotions.

Platform-wide AOV: Rs 1,371

Highest AOV category: Electronics (Rs 2,276)

Lowest AOV category: Grocery & Essentials (Rs 500)

8. Cities Generating the Highest Transaction Volumes

Bengaluru is the largest city-level contributor by transaction volume, with 1,615 transactions (10.47% of all recorded transactions). The distribution shows demand is fairly spread across the 14 major cities in the dataset, without extreme concentration in any single market.

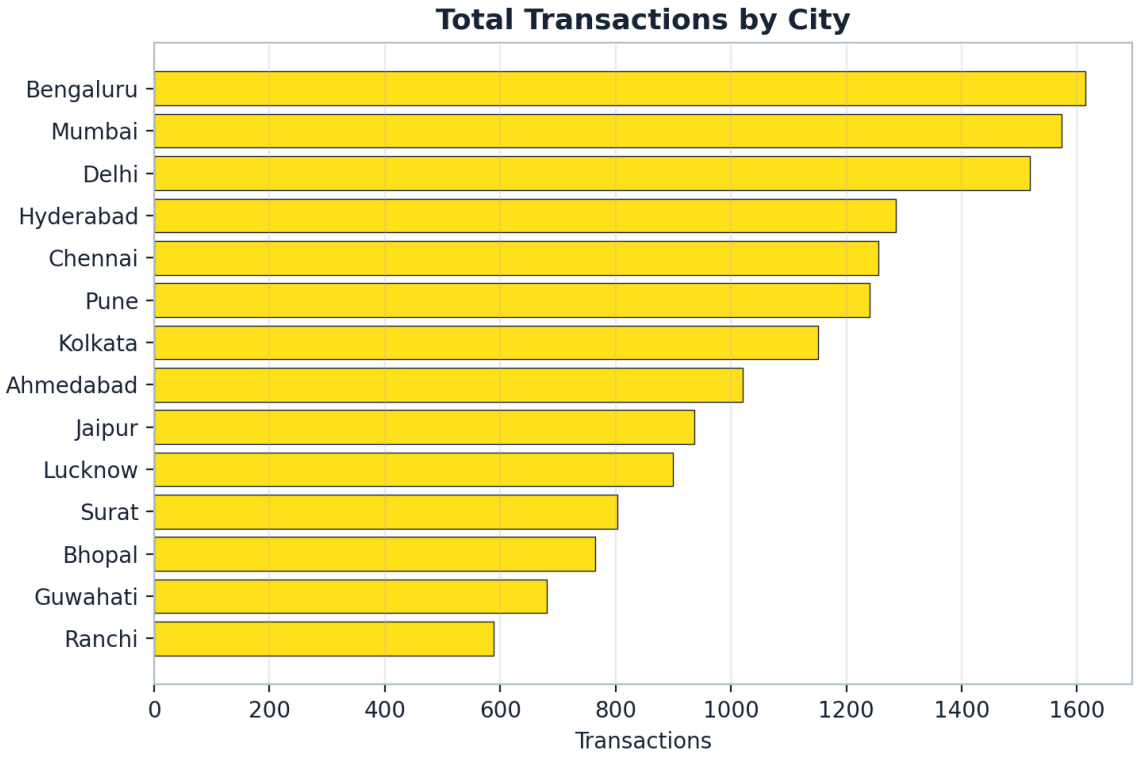


Figure 5. Total transactions by city

INSIGHTS:

- The top 6 cities (Bengaluru, Mumbai, Delhi, Hyderabad, Chennai, Pune) each record over 1,241 transactions.
- Bhopal, Guwahati, Ranchi sit at the lower end, each around 589–764 transactions.
- City-level transaction analysis can help with regional business planning.

Top-performing city: Bengaluru

Recorded transactions: 1,615

Share of total transactions: ~10.47%

9. Cities Contributing the Most GMV

Bengaluru is the largest contributor by GMV, at ~Rs 21.97 Lakh, leading the market alongside strong showings from the other top metros, reflecting consistently high average order values across major hubs.

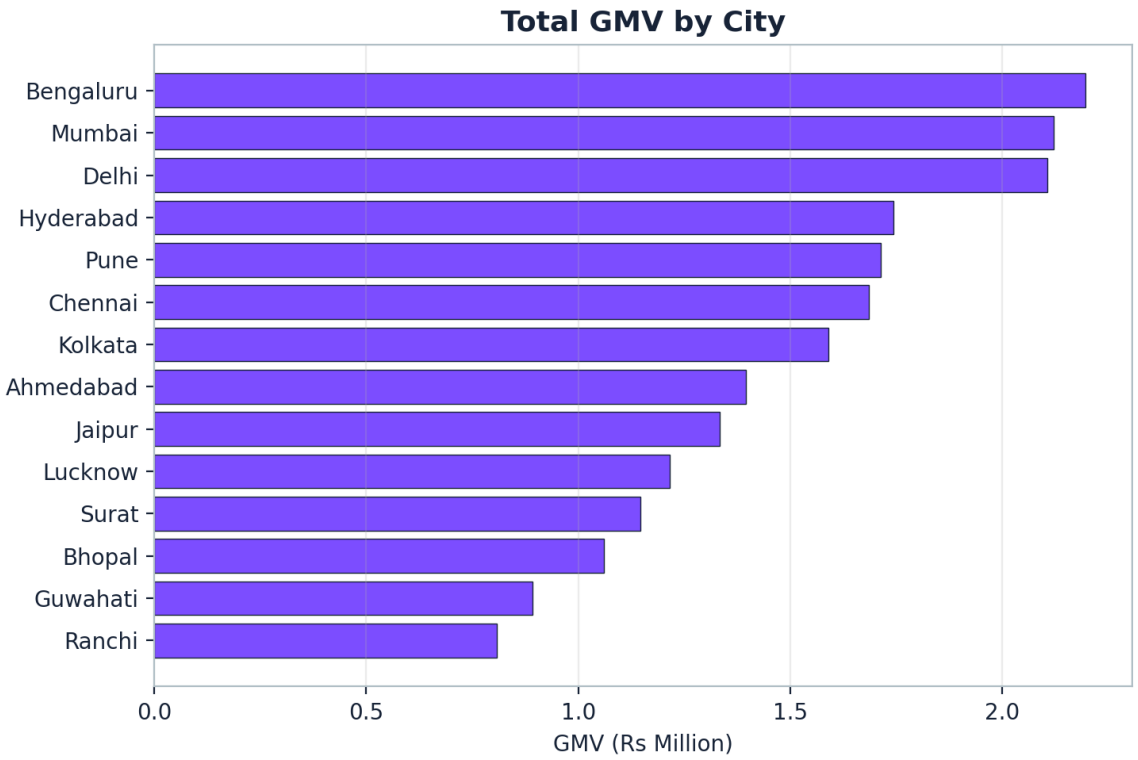


Figure 6. Total GMV by city

INSIGHTS:

- Bengaluru, Mumbai, Delhi, Hyderabad, Pune together account for the bulk of total GMV.
- GMV rankings closely mirror transaction-volume rankings, since average order values are fairly consistent across cities.
- High-GMV cities can be important markets for operational and service planning.

Top-performing city by GMV: Bengaluru

Recorded GMV: ~Rs 2,196,601

10. Average Discount Levels (Operational View)

The average discount applied across all transactions is 17.68%, and this varies noticeably by product category — fashion-led promotions and value-pricing on essentials shape a more segmented discounting strategy than a flat catalog-wide policy.

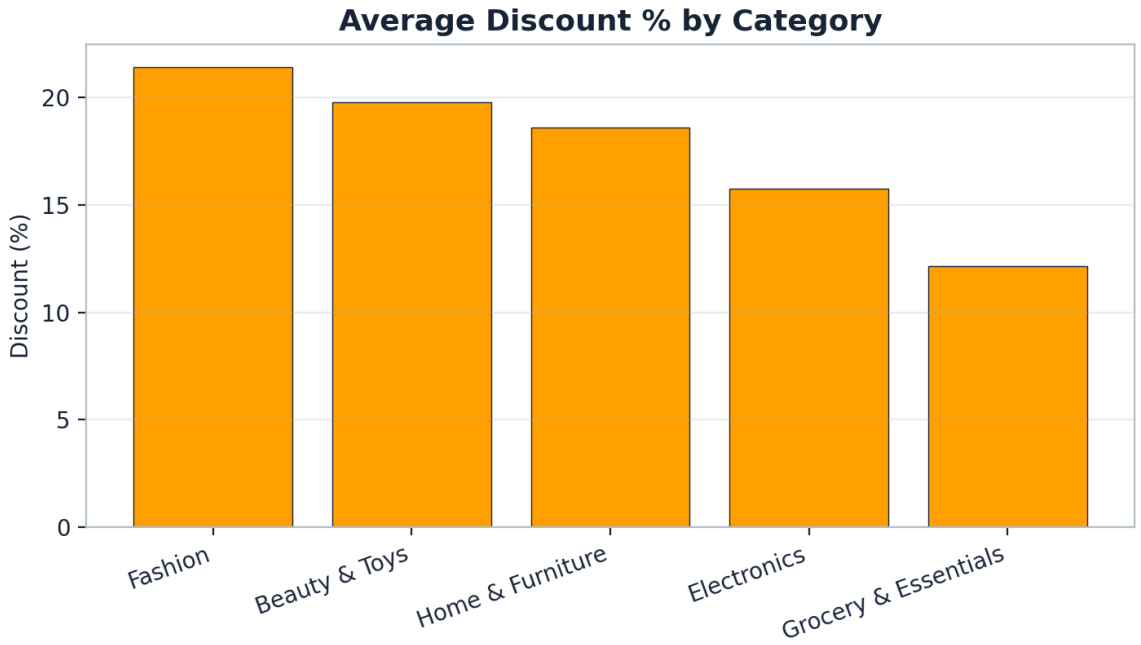


Figure 7. Average discount % by category

INSIGHTS:

- Fashion has the highest average discount (21.43%), reflecting frequent seasonal and end-of-line promotions.
- Grocery & Essentials has the lowest average discount (12.15%), consistent with tighter margins on everyday essentials.
- The spread across categories suggests a deliberately segmented discounting approach rather than a uniform catalog-wide policy.

Average discount (overall): 17.68%

Highest-discount category: Fashion (21.43%)

Lowest-discount category: Grocery & Essentials (12.15%)

11. Sale-Event Spike Analysis

Two Big Billion Days-style sale windows fall within the observation period (10–13 April and 18–21 May 2026). Isolating these 8 days from the remaining 82 regular days shows how sharply promotional events move platform demand.

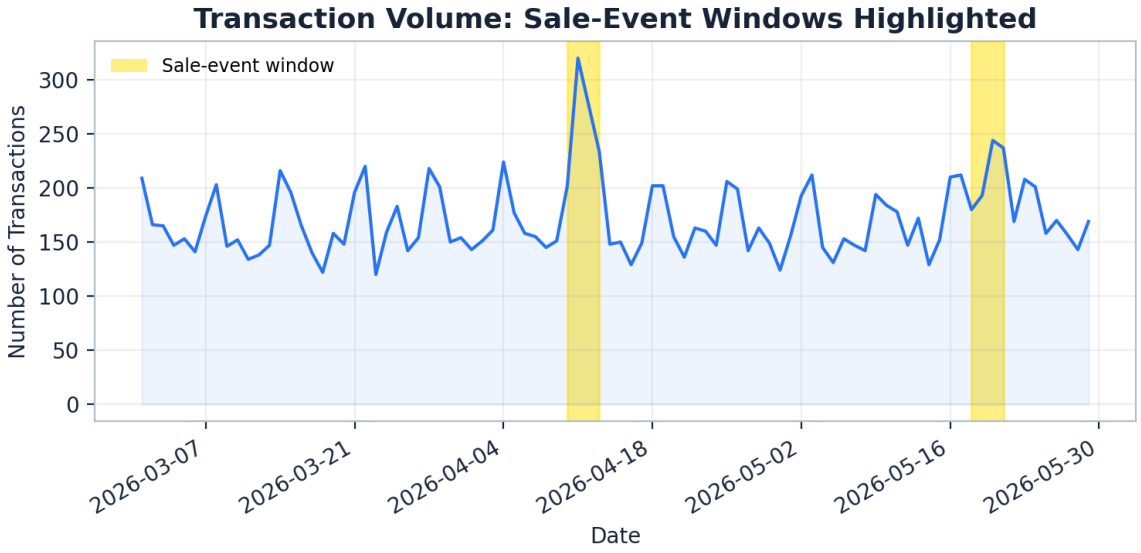


Figure 13. Transaction volume with sale-event windows highlighted

INSIGHTS:

- Average daily transactions jump to ~236 during sale windows versus ~165 on regular days — a 42.9% uplift.
- Just 12.2% of days in the window generate 12.0% of total GMV (Rs 0.25 Cr), underscoring how concentrated sale-event revenue is.
- Electronics sees the strongest pickup during sale windows, consistent with big-ticket electronics deals anchoring platform-wide promotions.
- This pattern supports dedicated sale-event capacity planning: inventory, logistics and customer support all need to flex sharply for a handful of high-intensity days.

Sale-event days in window: 8 of 90

Avg transactions/day (sale vs regular): ~236 vs ~165

GMV share from sale-event days: 12.0%

12. Regional Demand

Transaction demand is distributed across 14 major cities (excluding 90 rows with an unrecorded city). Bengaluru has the highest transaction volume in the supplied data at 1,615 transactions. Regional demand analysis can support fulfilment planning, city-level marketing spend, and service monitoring.

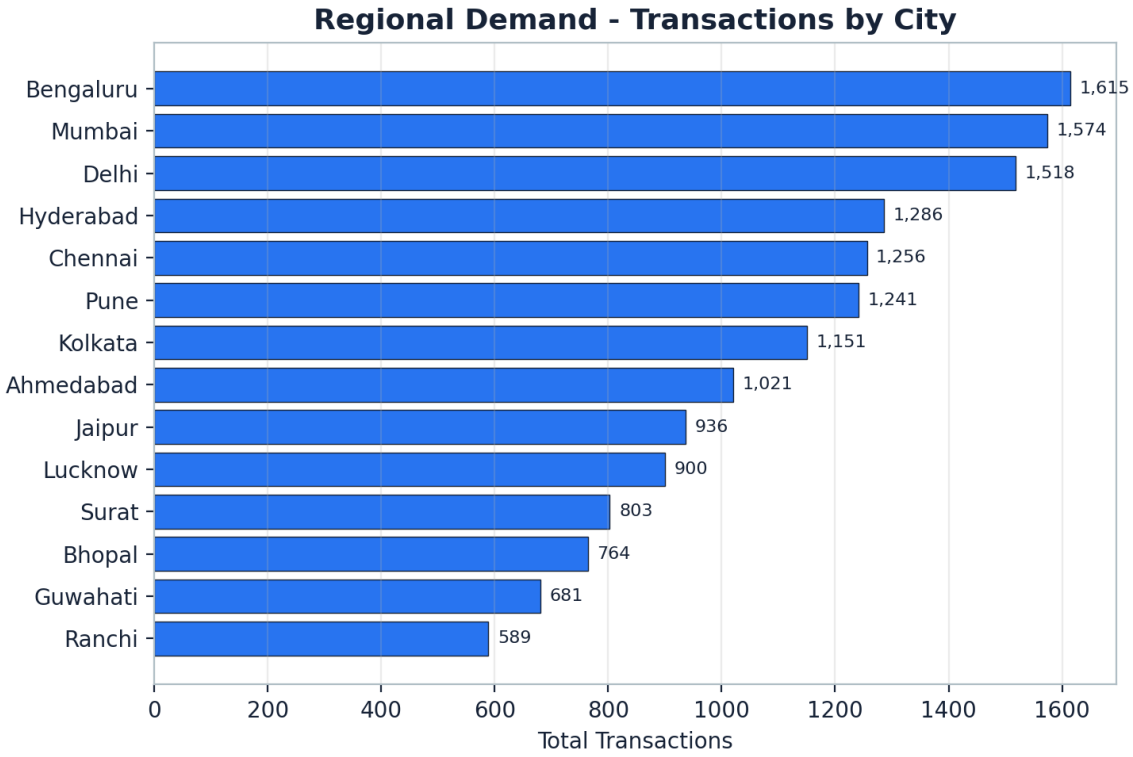


Figure 8. Regional demand — transactions by city

INSIGHTS:

- Demand is distributed across 14 cities without one dominant hub.
- Bengaluru represents the strongest demand market in the dataset, narrowly ahead of Mumbai.
- City-level demand information can support warehouse and delivery-fleet coverage decisions.

Top-performing city: Bengaluru

Transaction volume: 1,615

Cities covered: 14 (+90 records with unrecorded city)

13. Category x City Cross Analysis

Combining product category with city reveals where demand is most concentrated and highlights which local markets over- or under-index on specific categories relative to the national mix.

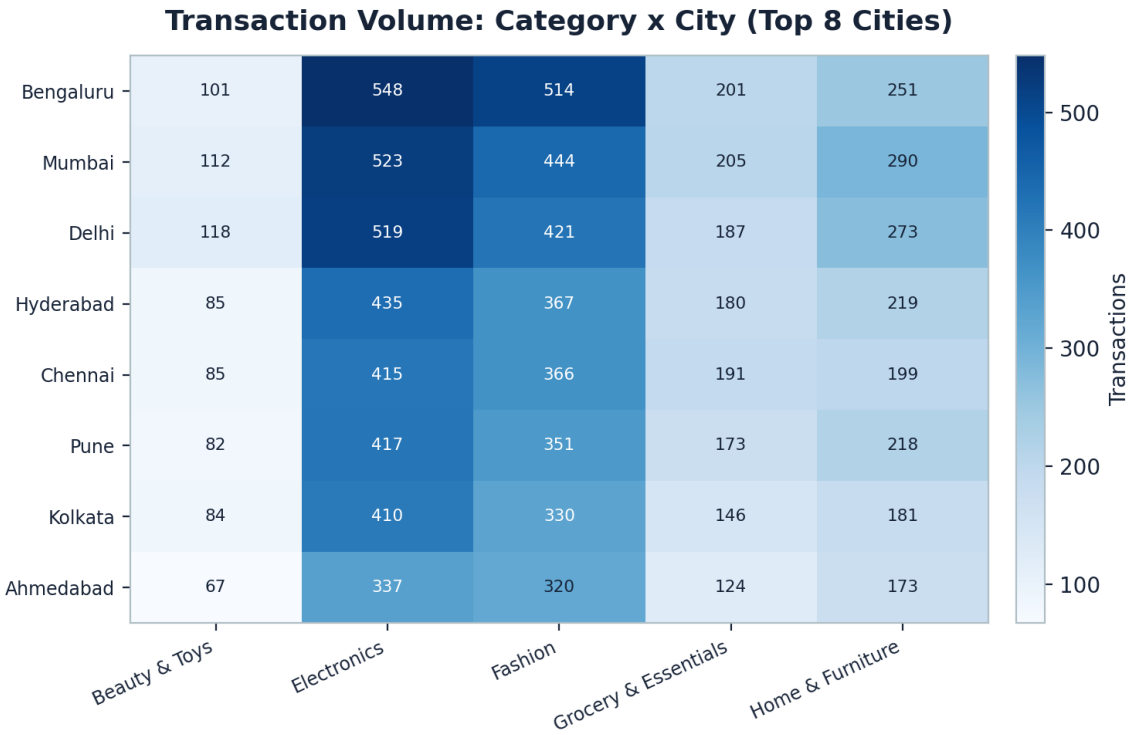


Figure 14. Transaction volume heatmap: category x city (top 8 cities)

INSIGHTS:

- The single strongest city-category combination is **Bengaluru + Electronics**, with 548 transactions — the darkest cell on the heatmap.
- Electronics and Fashion dominate every major city's mix, but their relative order shifts slightly city to city, hinting at local category preferences worth exploring further.
- Beauty & Toys stays the smallest column across all cities, confirming it is a genuinely niche category nationally rather than being suppressed in any one market.

Strongest city-category combination: Bengaluru + Electronics

Transactions in that combination: 548

Cities compared: Top 8 by transaction volume

14. Repeat-Customer Share vs Transaction Volume

The relationship between the share of repeat customers and transaction volume by city gives an additional descriptive view of city-level customer loyalty. Cities with a higher repeat-customer share do not necessarily have higher transaction volume, but the scatter should be read as association rather than causation.

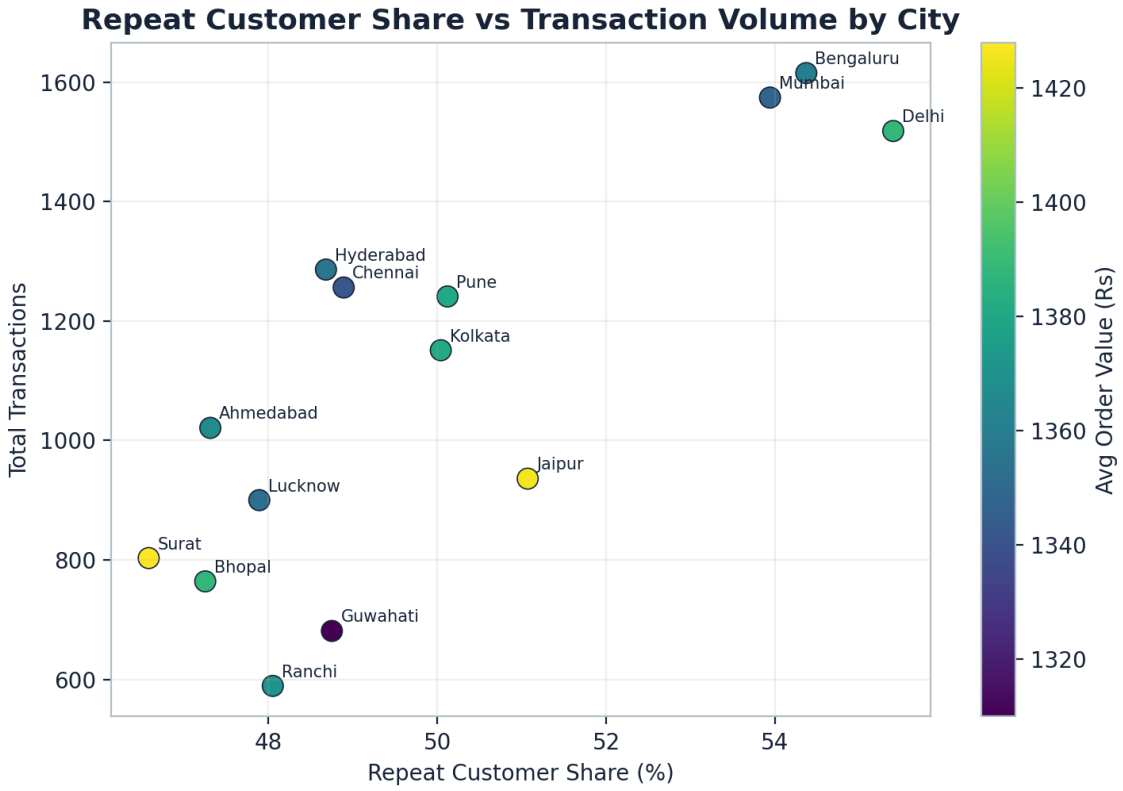


Figure 9. Repeat-customer share vs transaction volume by city

INSIGHTS:

- Repeat-customer share ranges between 46.58% and 55.4% across cities, showing moderately consistent loyalty patterns nationally.
- Delhi has the highest repeat-customer share (~55.4%), pointing to strong customer retention in that market.
- Average order value does not vary strongly with either transaction volume or repeat share.

Overall repeat-customer share: 50.57%

Highest repeat share: Delhi (~55.4%)

Lowest repeat share: Surat (~46.58%)

15. Weekday vs Weekend Transaction Pattern

Across the daily observations, average transactions are 156 per day on weekdays versus 210 per day on weekends — weekends run noticeably busier.

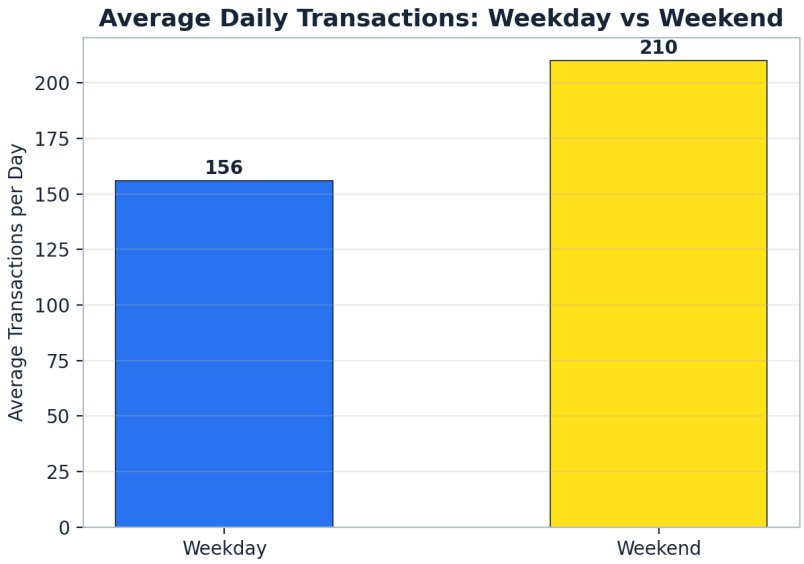


Figure 10. Average daily transactions: weekday vs weekend

INSIGHTS:

- Weekend transaction volume is about 34.6% higher than weekday volume on average.
- Average order value is only marginally higher on weekends (~Rs 1,379 vs ~Rs 1,367 on weekdays), so the weekend uplift is driven mainly by transaction count rather than basket size.
- This pattern can inform weekend-specific staffing, delivery capacity and promotional timing.

Average weekday transactions/day: ~156

Average weekend transactions/day: ~210

16. Customer Rating (CSAT Equivalent)

The overall average customer rating is 4.15 / 5. Bhopal has the highest average rating at 4.18, while Kolkata has the lowest at 4.03. Rating is not recorded for 1,659 transactions in the supplied data (~10.8%).

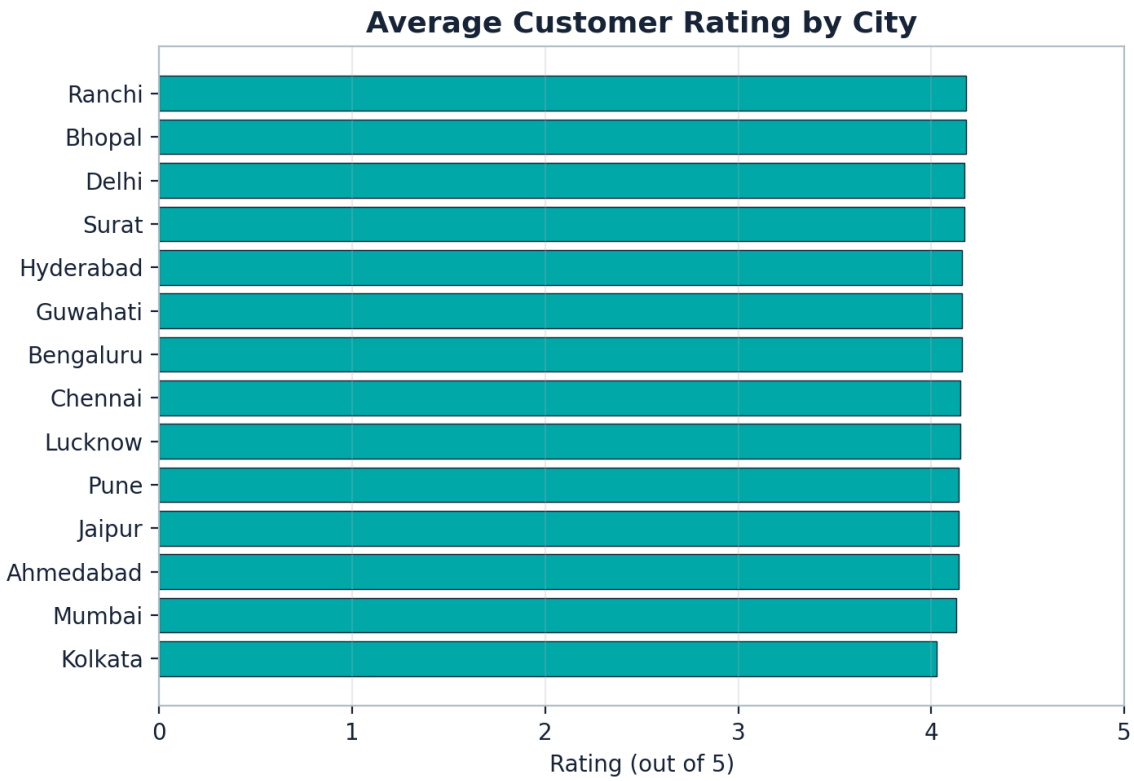


Figure 11. Average customer rating by city

INSIGHTS:

- Ratings are clustered between 4.03 and 4.18 across all cities, showing no city with a clearly poor customer experience.
- Kolkata's comparatively lower rating may warrant closer investigation into delivery or product-quality feedback.
- Rating should be considered alongside operational KPIs such as discount levels and repeat-customer share.

Highest-rated city: Bhopal (4.18/5)

Lowest-rated city: Kolkata (4.03/5)

Overall average rating: 4.15/5

17. Summary of Key Metrics

Metric	Value
Total transactions	15,425
Total GMV	Rs 2.11 Crore (Rs 21,144,007.16)
Platform-wide average order value	Rs 1,371
Average discount	17.68%
Average customer rating	4.15 / 5
Cities tracked	14 (+90 unrecorded)
Product categories tracked	5
Repeat-customer share	50.57%
Leading category by transactions	Electronics (34.11% share)
Leading category by GMV	Electronics (56.65% share)
Leading city by transactions	Bengaluru (1,615 transactions)
Leading city by GMV	Bengaluru (Rs 2,196,601)
Weekend transaction uplift	34.6% vs weekdays
Sale-event transaction uplift	42.9% vs regular days
Dataset records	15,425
Date range	1 Mar 2026 - 29 May 2026 (90 days)

18. Key Business Recommendations

Translating the descriptive findings above into concrete, actionable next steps for category, regional and customer-experience teams.

1. **Double down on Electronics**
   With 56.65% of GMV from 34.11% of transactions, Electronics is the single highest-leverage category for revenue growth. Prioritise inventory depth, seller onboarding and financing offers here.

2. **Build sale-event playbooks**
   Sale windows deliver a 42.9% demand spike and 12.0% of GMV in just 12.2% of days. Pre-stage logistics, customer support and server capacity around these recurring windows rather than treating them as one-off events.

3. **Grow basket size in low-AOV categories**
   Grocery & Essentials has the lowest AOV at Rs 500. Bundle offers and cross-sell prompts alongside high-frequency essentials purchases can lift basket value without hurting conversion.

4. **Localise city-category promotions**
   Bengaluru + Electronics is the strongest single combination in the data. Use the category x city cross-analysis to target city-specific promotions rather than running identical national campaigns everywhere.

5. **Investigate lower-rated markets**
   Kolkata trails on customer rating (4.03/5). A focused review of delivery timelines and product-quality feedback in this city can lift CSAT with a targeted, low-cost intervention.

6. **Plan weekend-specific operations**
   Weekend volume runs 34.6% above weekday levels with flat basket size. Align staffing, delivery slots and weekend-only promotions to this predictable rhythm.

19. Next Steps to Move from Descriptive → Predictive

- Demand forecasting by city and category to estimate future transaction volumes.
- Anomaly detection for sudden changes in discount levels, GMV or transaction volume.
- Churn analysis linking repeat-customer share to marketing and retention spend.
- Basket-level analysis using average order value and quantity per transaction.
- Trend and cohort analysis when a longer historical window becomes available, including sale-event (Big Billion Days-style) impact modelling.

This report answers what happened in the supplied dataset. The next analytical layer would use a longer time period and richer event-level data to explain why changes occurred and to forecast what is likely to happen next.

20. Glossary

I. GMV — Gross Merchandise Value: Total monetary value of orders placed through a platform before deducting fees, discounts already applied, refunds, or other adjustments. Here it is calculated as the sum of order_value_inr.

II. MRP — Maximum Retail Price: The listed price of a product before any discount is applied.

III. City Tier — A classification of Indian cities by size and economic activity (e.g., Tier 1, Tier 2/3), commonly used for regional business planning.

IV. Repeat vs New Customer — A flag indicating whether a transaction was placed by a customer with a prior purchase history (Repeat) or a first-time buyer (New).

V. Rating — A post-purchase customer rating (out of 5), used here as a proxy for customer satisfaction (CSAT).

VI. AOV — Average Order Value: The mean value of a transaction within a given category, city, or time period, calculated as total GMV divided by transaction count for that slice.

VII. Sale-Event Window — A short, promotion-driven period (e.g., a Big Billion Days-style sale) during which transaction volume and GMV spike well above baseline levels.

VIII. Cross-Analysis — A descriptive technique that combines two dimensions (e.g., category and city) to reveal patterns not visible when each dimension is viewed alone.

IX. CSAT — Customer Satisfaction: A metric capturing how satisfied customers are with their purchase experience, approximated here using the post-purchase rating field.

THANK YOU